

Sharing and Experiencing Hardware and Methods to Advance Smell, Taste, and Temperature Interfaces

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ABSTRACT

There has been a monumental push from the CHI community to bring more human senses to interactive devices. This trend is significant because we use all our senses in everyday interactions but only an *extremely narrow subset* when interacting with computers. This workshop focuses on bringing together researchers to advance some of the most challenging senses to embed into interfaces, but arguably the most exciting: smell, taste, and temperature. To integrate these modalities into interfaces, researchers not only use methods from traditional mechanics or haptics (e.g., pumps, heating pads, etc.) but must also acquire tacit skills and understandings from psychophysics, neuroscience, anatomy, and chemistry (e.g., receptor signaling pathways or food chemistry). This demo-based workshop provides a platform to come together and bring their demonstrations, experiments, and hardware to experience, discuss, and advance the field.

CCS CONCEPTS

• **Hardware** → Emerging technologies; Emerging interfaces; • **Human-centered computing** → Human computer interaction (HCI).

KEYWORDS

Smell, taste, temperature, chemesthesis, demonstrations

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1 BACKGROUND

In the past decades, there has been a monumental and multi-disciplinary effort from all corners of Human-Computer Interaction to bring more human senses into interactive systems [4]. This trend is significant because we use *all our senses* in critical, everyday interactions, yet only an *extremely narrow subset* when interacting with computers. Many sub-fields of HCI note this discrepancy. For example, in virtual reality, deserts and mountains do not feel real if not warm and cold [5] or without scent [2, 3], or in health, we need new, visceral ways to intervene in the ways we eat [9, 13], train our senses [10, 11], or engage with our emotions [2] and sleep [1].

We particularly focus on advancing three of the most challenging and exciting senses to embed into interfaces: **smell, taste, and temperature**. Implementing these senses into interfaces is notoriously complex from both a *technical* and a *user-experience* standpoint. Researchers not only engineer these devices with techniques inherited from other fields (e.g., pumps and heating pads from haptics or robotics) but also must integrate insights from neuroscience, biology, and chemistry (e.g., receptor signaling pathways, food chemistry, cross-modal interactions). Take, for example, an HCI researcher interested in *incorporating the sense of taste* into their interface to increase the embodiment of a cooking simulator. Our researcher must (1) engineer a taste stimulator, (2) gather insights from chemistry while designing the stimuli (which chemicals create the simulated tastes and how they combine well), (3) integrate insights from neuroscience on how taste interacts with existing smells in the user's environment, (4) understand the rate at which humans swallow and perceive taste changes, and (5) more. The complexity of these interfaces is tremendous.

One way to **accelerate the research** in this space is by **bringing researchers together to share their techniques, findings, tacit knowledge, and even hardware**. As such, this workshop aims to provide a platform for researchers to gather and provide demonstrations, experiments, and hardware for others to experience. This experience- or demo-based format will collaboratively instigate discussions and advance the field of smell, taste, and temperature beyond its current from-scratch methods. Figure 1 depicts four exemplary hardware devices (from the organizers) that illustrate the type of hardware our participants will bring to the table for open discussion and hands-on experience.



Figure 1: Examples of smell, taste, and temperature hardware: (a) *The Smell Engine* [3]; (b) *Chewing Jockey* [13, 14]; (c) *temperature altering perceived softness* [12]; and (d) *stereo-smell sensing via electrical trigeminal stimulation* [6].

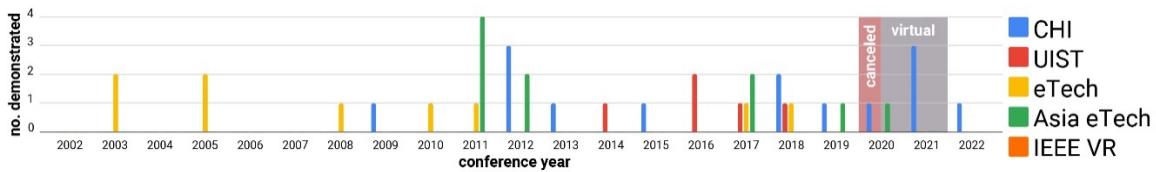


Figure 2: Trends for the number of demonstrated smell, taste, and temperature interfaces over the last two decades. Despite publishing numbers exploding in the same time frame, firsthand opportunities to experience this research remain scarce and fragmented, with the most (4) interfaces demonstrated at SIGGRAPH Asia eTech 2011 and CHI 2012 or 2021 (virtual) close behind.

2 A CALL TO DIRECTLY EXPERIENCE SMELL, TASTE, AND TEMPERATURE

Despite the increase in publications on smell, taste, and temperature [4], researchers continue to develop their complete hardware and experiments from scratch. This repeated labor happens because we never had a platform to bring our devices to the table and share our findings with one another from the angle of **democratizing research and research tools for smell, taste, and temperature interfaces**. This angle is the call to action of our workshop.

Unlike traditional interfaces based on vision or audition, chemo- and thermo-sensory experiences *must be experienced firsthand*. Their actual experience is often difficult to communicate via words. In other words, our papers do not perfectly describe to other researchers how our interfaces *feel*. Figure 2 illustrates this gap between *published and demonstrated works*, including aggregated data from several conference demonstration sessions, notably CHI, UIST, SIGGRAPH eTech, SIGGRAPH Asia eTech, and IEEE VR.

As shown, opportunities to directly experience smell, taste, temperature, and chemesthetic works remain rare. Moreover, as new researchers enter the field, most never get to experience the past projects that have shaped the field (e.g., *Honey I'm Home* [8] or *Food Simulator* [7]). Through the workshop, we hope to address this gap by **providing a space for researchers to share and experience their hardware, experiences, and experimental setups**.

3 WORKSHOP

3.1 Topics of Interest

Smell, Taste and Temperature interfaces have unique challenges and opportunities, such as the challenge of developing suitable and robust hardware, on top of implications and open questions

in biology and neuroscience (e.g., in smell research, multimodal integration of smell/taste and much more). The topics of interest for the workshop include, but are not limited to, the following:

• **Sharing & open-sourcing experimental setups, hardware, or toolkits for smell, taste, and temperature.**

- Techniques and methods to design and use multi-sensory technologies.
- Systems for recording, stimulating, and augmenting multi-sensory experiences.
- Assistive and environmental technologies using smell, taste, or temperature.
- Cultural aspects and contexts of multi-sensory interactions in everyday life and history, and how they influence and shape the technologies developed and their societal adoption.

We aim to additionally highlight and discuss open challenges in the field, which include, but are not limited to,

- Recommendations on sharing tacit knowledge.
- **Sharing system designs:** share how researchers overcome power limitations, wearable form factor constraints, network constraints, and compute resource utilization.
- **Tackling the replication of this research:** affects all these devices, which end up being often non-reducible, recordings and replay of experiences, precision, inter-, and intra-modality effects.
- **Data sharing & modeling:** how to share data across HCI researchers, benefit from recommended practices in data modeling, information dissemination, and evaluation methodologies.

- **Perception:** avoid sensory conflicts or take advantage of mixed senses, calibrate for over or under stimulation, predictably stimulate senses for a large and diverse population of people.
- **Social implications:** for example, questions around cultural adoption and equal access to these interfaces.

3.2 Goals and Rationale

This year, the workshop's goal is to create an environment where relevant researchers and practitioners can directly experience each other's work, hardware, or experiments through "demonstrations" and "insights sessions." In a demonstration, the participants show their technical setup, experiment, or experience and let others explore it in smaller groups, lending itself well for small breakout discussions. Alternatively, in an insights session, a participant discusses and dives deep into a particular insight they gained in their work (e.g., sharing tacit knowledge): for example, an experimental design, a finding, or even recommendations based on open-source devices they engineered.

We look forward to fostering a thriving community of researchers, practitioners, and artists conducting research at the intersection of technology and chemo- or thermal sensations (smell, taste, temperature). The workshop will instigate conversations about recent trends or challenges and catalyze new ideas for future multi-sensory technologies and events.

In 2020, we self-organized an online symposium that served as the first pilot for our proposal. This symposium included many papers that were not shown due to the cancellation of CHI 2020. We had 255 people registered from over 33 countries worldwide. At CHI 2021, we organized a virtual workshop that was the first time researchers had a forum to present their work. Finally, in this proposal, we break away from the traditional talk-based workshop to provide a platform for researchers to share their hardware, findings, and tacit knowledge in a hands-on and collaborative manner to advance the field.

3.3 Organizers

We provide short biographies of each workshop organizer. The organizers' experiences are complementary, providing a wide range of perspectives on the workshop's topic.

Jas Brooks is a Ph.D. student in the Department of Computer Science at the University of Chicago. Their research expands the computer interface to engage with the chemical interactions of our bodies and investigates what this paradigm enables. These chemical interfaces have introduced new input and output methods, such as electrically induced stereo-smell (directional odor cues), chemically stimulated haptics, and more. Their research is covered in media publications like WIRED, Fast Company, Digital Trends, and IEEE Spectrum, and supported by a National Science Foundation Graduate Research Fellowship. Website: jasbrooks.net

Alireza Bahremand is a Ph.D. student in the School of Electrical, Computer, and Energy Engineering at Arizona State University. His research focuses on adaptive software-hardware frameworks for multi-sensory digital embodiment. Recent works include the

Smell Engine, a software-hardware framework that integrates olfactory stimuli into virtual environments, closely reflecting real-world olfactory sensations. As a contributing researcher to the ASU Dreamscape Learn partnership, Alireza is working with teams of professors and students to build educational multi-sensory XR experiences at scale. Website: alirezabahremand.com

Pedro Lopes is an Assistant Professor in Computer Science at the University of Chicago. Pedro focuses on integrating computer interfaces with the human body—exploring the interface paradigm that supersedes wearable computing. Some of these new integrated devices include: a device based on muscle stimulation that allows users to manipulate tools they never seen before or that accelerate their reaction time, or a device that leverages the nose to create an illusion of temperature. Pedro's work also captured the interest of media, such as New York Times or NewScientist, and was exhibited at Ars Electronica and the World Economic Forum. Website: lab.plopes.org

Christy Spackman is an Assistant Professor in the School for the Future of Innovation in Society as well as the Department of Arts, Media, and Engineering at Arizona State University. Her work focuses on how the sensory experiences of making, consuming, and disposing of food influence and are influenced by "technologies of taste," oft-overlooked technologies and practices used to manage the sensory aspects of foods during production. Her current book uses this framework to examine how scientific and technological innovation changed the taste of water throughout the 20th-century. Christy holds training in molecular biology, food chemistry, and the culinary arts, allowing her to bridge disciplinary divides in research. Website: christyspackman.com

Judith Amores is a Research Fellow at the MGH/Harvard Medical School and a Research Affiliate at the MIT Media Lab, where she did her Ph.D. and master's and helped run VR/AR at MIT as a co-president. She holds a multimedia engineering degree and has worked at Microsoft Research, URL Barcelona, and the Google Creative Lab. Her awards and publications include over 27 peer-reviewed research papers, two patents, a Facebook Graduate Fellowship, LEGO Foundation sponsored research, and was a finalist of the Innovation by Design Awards. She also received the Scent Innovator Award by CEW and IFF. Website: judithamores.com

Hsin-Ni Ho is an Associate Professor in the Faculty of Design at Kyushu University, Japan. Her research interests include haptic perception, multimodal interactions, shitsukan (material quality) perception, and the development of haptic interfaces based on human perceptual properties. Hsin-Ni received her Ph.D. in Mechanical Engineering from the Massachusetts Institute of Technology in 2006 and worked as a research scientist in the NTT Communication Science Laboratories, Japan, from 2007 to 2021. Website: sites.google.com/view/hohapticslab

Masahiko Inami is a Professor at the University of Tokyo whose research asks, "What are the challenges in creating interfaces that allow users to act and express their intentions intuitively?" Today's computer systems are limited: they only exploit visual and auditory sensations. However, daily life uses a variety of modalities, including some that come in direct contact with our bodies. These later modalities can dramatically affect our ability to experience and express ourselves. Masahiko designs a new generation of Human-Computer Integrated systems informed by modern physiological

understandings of sensation and perception, emerging electronic devices, and agile computational methods. Website: star.rcast.u-tokyo.ac.jp

Simon Niedenthal is an Associate Professor of Interaction Design at Malmö University, Sweden. His research focuses upon olfactory interaction, game aesthetics, and the sensory dimensions of gameplay. Recent projects include *Nosewise*, a collaboration with psychologists at Stockholm University exploring the potentials of smell training, and *Exerscent*, an open source, low-cost olfactory interface for remote olfactory assessment and training. In 2008, Simon defended his Ph.D. thesis *Complicated Shadows: The Aesthetic Significance of Simulated Illumination in Digital Games*, on game lighting and its effect on the emotions and behavior of the player. Website: Google Scholar

We are additionally receiving logistical and planning support from Jessica Lai, Mason Manetta, and Lauryn Mannigel.

Jessica Lai is a Ph.D. student in the Department of Engineering and Design and the Department of Arts at Arizona State University. Her interests lie at the intersection of electrical engineering and visual arts, focusing on human-computer interaction, virtual reality, and olfaction. Her most recent work was a flexible software-hardware infrastructure for producing spatially sensitive and programmable odors. She is working to further her insights in this realm while developing an artistic and scientific virtual reality experience about the senses through a journey to Mars.

Mason Manetta is an Electrical Engineering undergraduate researcher at Arizona State University's Meteor Studio. Mason focuses on researching and developing odor-producing computer interfaces for multimodal experiences. Their most recent device enabled spatial smell perception in virtual reality. Additionally, Mason investigates energy-efficient architectures for mobile and embedded image capture and medical imaging methods, particularly electrical impedance tomography. Mason has published work at IEEE VR and MobiSys, which have been covered in media articles by ASU and KJZZ public radio. Mason plans to pursue a doctoral degree. Website: mcmannetta.github.io

Lauryn Mannigel is an artist-researcher and curator based in Tempe (U.S.) and Berlin (D.E.). She is a Ph.D. student in the Media Arts and Sciences program at Arizona State University. Since 2005, Mannigel has been tackling the issue of the Western cultural dominance of visual aesthetics and epistemology by exploring primarily non-visual sensory perception through diverse media technologies. Since 2015, Mannigel's primary research interest has focused on the political potential of smell. She exhibited artwork and spoke at conferences worldwide. Website: laurymannigel.me

3.4 Structure

The two-day workshop will be organized in an in-person format. Our workshop consists of a demo session, panels, provocation talks to spark discussion, and in-depth breakout discussions. All in-person talks will be recorded. All videos will additionally be closed captioned for accessibility purposes prior to public release. We expect 20-25 accepted submissions and 20-40 total participants (not counting organizers). Based on past iterations, we expect to fill these seats. To additionally promote interaction before, during,

and after the workshop, we continue communications on the Discord Server started in 2021 for text-based communication. Beyond the insight sessions and provocations, we will host demonstration sessions so participants can interact with and experience research projects and related art.

3.4.1 Website. The workshop webpage can be found at smell-tastetemperature.com. The website will promote the workshop theme, and will include background information on the topic, Call for Participation, workshop schedule, backgrounds of each of the organizers, and links to relevant papers and news. Additionally, accepted applications will be publicly available for download alongside, later, the paper presentation and demonstration recordings.

3.4.2 Pre-Workshop Plans. We will circulate the workshop's Call for Participation via mailing lists (e.g., ACM SIGCHI, UIST) to researchers in the field of HCI, UX, and related areas interested in multisensory research. We will also send it to all participants of our previous self-organized workshops (250+ participants) as well as to relevant arts and humanities organizations (e.g., the Institute for Art and Olfaction, Aroma Discord, the Association for the Study of Food and Society, Olfactory Art Keller, Mediamatic). Demonstration and talk recordings will be made available on the STT'23 website.

To summarize important dates for the workshop, we gather them below:

- **Final submission deadline:** Friday, February 24, 2023, at 12:00pm (noon) PT.
- **Notification date:** Friday, March 10, 2023, at 12:00pm (noon) PT.
- **Video deadline:** Friday, April 7, 2023, at 12:00pm (noon) PT.

3.4.3 Activities and Schedule. The one-day workshop is split into three brainstorming featuring provocation talks by organizers and invited speakers followed by discussions all held in-person and over Zoom. Provocation talks are short presentations meant to reframe the current trends or future possibilities with an organizer's evocative vision. The exact times of these brainstorming will be scheduled according to participants, in consideration of varying time zones.

Day 1 – Challenges & Social Impact – Start at 9:00 AM

- Welcome by Jas, Alireza, Jessica, and Mason (15m)
 - Introductory hands-on provocation: against blank-slate futures.
- Social icebreaker (30m)
- **Panel 1** Sharing hardware & democratizing research on smell, taste & temperature (1h30m)
 - 30m Provocation panel (panelists to be determined)
 - 10m Q&A
 - 20m Attendee presentations
 - 20m Moderated discussion with presenters
- Bathroom and set-up break (15m)
- **Demonstration sessions** (all participants in small groups, 2h)
- Lunch break (1h)
- **Session 2** – Societal implications for Smell, Taste & Temperature Interfaces (1h30m)
 - 30m Provocation by Prof. Christy Spackman (ASU)
 - 10m Q&A
 - 20m Attendee presentations

- 20m Moderated discussion with presenters
- Bathroom break (15m)
- Group activity (1h)
- Hotel/relaxation break (2h)
- Optional food event led by Mason and Jessica.
- (Potentially dinner with beer tour or a rose garden smell walk followed by dinner.)
- Day 2 – Designing the Future Interfaces *Together* – Start at 10:00 AM
- Breakfast activity/warm-up (20m)
- **Insights Session 3** – Sharing Techniques to Design & Open-Source Smell interfaces (1h30m)
 - 20m Attendee insight presentations
 - 20m Moderated discussion with presenters
 - 10m Short smell activity (break)
 - 20m Brainstorms
 - 20m Discuss ideas as workshop
- Lunch break (1h)
- **Insights Session 4** – Sharing Techniques to Design & Open-Source Taste interfaces (1h30m)
 - 20m Attendee insight presentations
 - 20m Moderated discussion with presenters
 - 10m Short taste activity (break)
 - 20m Brainstorms
 - 20m Discuss ideas as workshop
- Bathroom break (30m)
- **Insights Session 5** – Sharing Techniques to Design & Open-Source Thermal/Chemical interfaces (1h30m)
 - 20m Attendee insight presentations
 - 20m Moderated discussion with presenters
 - 10m Short thermal and chemesthetic activity (break)
 - 20m Brainstorms
 - 20m Discuss ideas as workshop
- Bathroom break (30m)
- **Panel 2 (everyone):** Next steps as community & closing remarks (30m).

3.4.4 Post-Workshop Plans. All accepted submission papers will be made public on the website. We will additionally record all presentations, provocation talks, and demonstrations at the workshop to make them accessible via YouTube after the event finishes. The outcomes of 2023 (proposed) will be submitted as a summary report to ACM Interactions to reach out to a larger community of academics and practitioners interested in new frontier research in HCI. To additionally encourage interest, we will share outcomes and updates across online communities, such as HCI Facebook groups (e.g., CHI Meta, MPHCI&UX, ACM SIGCHI), media art groups (e.g., The New Media Thing), and multisensory-specific groups (e.g., Aroma Discord, the Institute for Art and Olfaction, the Association for the Study of Food and Society, Olfactory Art Keller, Mediamatic). Participants are encouraged to keep using the Discord Server even after the workshop.

4 CALL FOR PARTICIPATION

The “Experiencing Smell, Taste, and Temperature Interfaces” workshop addresses the burgeoning subfield of chemo- and thermo-sensory interfaces (smell, taste, and temperature) as well as their

cultural contexts, usage, and resulting experiences. This two-day, in-person workshop will offer an interdisciplinary forum of discussion for academics and practitioners interested in leveraging these sensations. The workshop especially provides space for researchers and practitioners to directly experience past and new interfaces through in-person demonstrations.

The topics of interest for the workshop include, but are not limited to, the following:

- **Sharing & open-sourcing experimental setups, hardware, or toolkits for smell, taste, and temperature.**
- Techniques and methods to design and use multi-sensory technologies.
- Systems for recording, stimulating, and augmenting multi-sensory experiences.
- Assistive and environmental technologies using smell, taste, or temperature.
- Cultural aspects and contexts of multi-sensory interactions in everyday life and history, and how they influence and shape the technologies developed and their societal adoption.
- We aim to additionally highlight and discuss open challenges in the field, which include, but are not limited to,
- Recommendations on sharing tacit knowledge.
- **Sharing system designs:** share how researchers overcome power limitations, wearable form factor constraints, network constraints, and compute resource utilization.
- **Tackling the replication of this research:** affects all these devices, which end up being often non-reducible, recordings and replay of experiences, precision, inter-, and intra-modality effects.
- **Data sharing & modeling:** how to share data across HCI researchers, benefit from recommended practices in data modeling, information dissemination, and evaluation methodologies.
- **Perception:** avoid sensory conflicts or take advantage of mixed senses, calibrate for over or under stimulation, predictably stimulate senses for a large and diverse population of people.

Researchers and practitioners from academia, arts, and industry are invited to apply to the workshop by submitting a 1-4 page paper or demonstration description in the ACM Master Article Submission Template single column format via the Easy Chair system (see workshop webpage). This paper should describe the demonstration or insight to be shared with the other attendees. The submission deadline is Friday, February 24, 2023, at 12:00pm (noon) PT. All applications will be reviewed by the workshop organizers and selection will be based on the paper’s quality. Applicants will be notified of decisions on Friday, March 10, 2023, at 12:00pm (noon) PT. Accepted papers and demonstrations will be made available at the workshop webpage. Upon acceptance, participants are asked to give a 5-minute presentation talk at the event and, if specifically applied for, a demonstration of their work. At least one author of each accepted paper must attend the workshop, and all authors are invited to the workshop Discord Server. All participants must register for both the workshop and for at least one day of the conference.

Further details can be found on the workshop webpage: <https://smelltastetemperature.com>

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